

# Longitudinal Analysis of Sepsis-3 Onset in the ICU: Toward Real-Time Prediction (MIMIC-IV)

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## 1 Introduction

A companion cross-sectional report analysed each ICU stay as a **single row** of first-24h summaries and asked “*was this stay septic?*” That framing cannot serve the project’s actual objective — a **real-time early-warning system** that, at each hour and using only the data observed so far, estimates the risk that a patient will develop sepsis. This report rebuilds the problem on its proper, **longitudinal** footing.

From the raw tables a pipeline (scripts 01–08) resamples every stay onto an **hourly grid** — a panel of **3,095,002** stay-hours across **65,241** stays (mean 47 h/stay). SOFA is scored every hour, and the hour of **sepsis onset** is located as the first hour, inside the suspected-infection window, at which SOFA reaches the organ-dysfunction threshold. This gives what the static matrix never had: a **time axis** and an **onset time**.

Two facts shape everything that follows:

1. **32.0%** of stays meet Sepsis-3 within the 72-hour horizon, but of these **34%** are already septic at ICU admission (onset hour 0) — *prevalent*, not predictable. The median onset is hour **2**.
2. The clinically useful question is therefore about **incident** sepsis: among patients *not yet* septic, who will become so? We answer it with a **landmark design** at hour **6** — conditioning on the first six hours and predicting onset thereafter — which is exactly how a deployed model would be used.

The methods are the longitudinal counterparts of the cross-sectional report: **Kaplan–Meier** and **Cox** (including a **time-varying Cox**) for time-to-onset, **Fine–Gray competing risks** (death competes with onset), **autocorrelation and trajectory** analysis, a **within-patient hidden Markov model** (the HMM applied to real sequences, not stay order), and **time-dependent discrimination** benchmarked against the streaming clinical scores a new model must beat.

## 2 The Hourly Panel and Onset Labelling

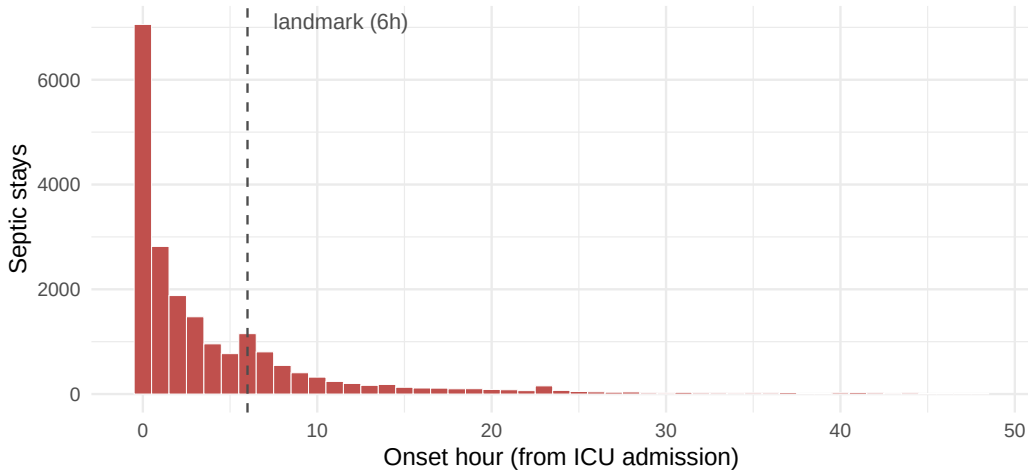


Figure 1: Distribution of sepsis-onset hour among septic stays. A large spike at hour 0 marks sepsis already present at admission (prevalent); the right tail is incident sepsis developing in the ICU — the target of early prediction.

Figure 1 shows the split between prevalent and incident sepsis. Conditioning at the hour-6 landmark leaves **48,829** stays still at risk, of whom **4,750** (9.7%) later develop sepsis and **3,330** (6.8%) within the next 12 hours. This is a realistically **imbalanced** early-prediction problem, unlike the near-balanced stay-level label.

## 3 Time-to-Onset: Kaplan–Meier

Time-to-event methods treat onset as an event that may or may not have happened by the end of observation (censoring), which a simple classifier ignores. The survivor function  $S(t)$  is the probability of remaining sepsis-free to hour  $t$ ;  $1 - S(t)$  is the cumulative incidence.

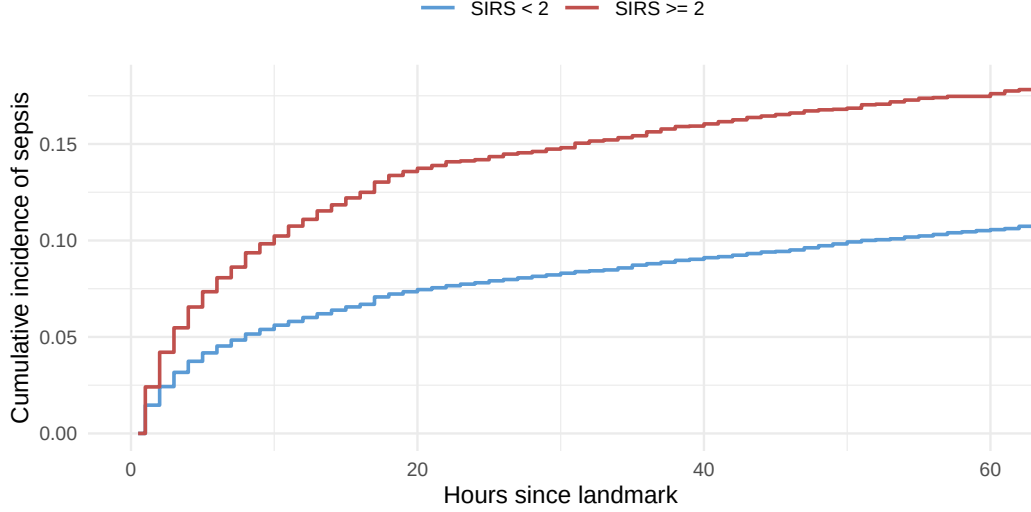


Figure 2: Cumulative incidence of sepsis from the hour-6 landmark, stratified by SIRS at the landmark. Patients meeting  $\geq 2$  SIRS criteria reach sepsis substantially faster.

Figure 2 shows the cumulative incidence separating early by SIRS status: by 24 h after the landmark it reaches roughly **14.2%** in the  $\text{SIRS} \geq 2$  group versus **7.8%** otherwise. A log-rank test confirms the difference.

The log-rank statistic is **360** ( $p < 1\text{e-}16$ ).

## 4 Cox Regression and the Failure of Proportional Hazards

A Cox model estimates how baseline (hour-6) covariates scale the hazard of onset. Hazard ratios (HR) above 1 accelerate onset.

Table 1: Cox model for time to incident sepsis (hour-6 landmark).

| Term       | HR    | 95% CI         | p                |
|------------|-------|----------------|------------------|
| age        | 1.009 | [1.007, 1.011] | $< 1\text{e-}16$ |
| hr         | 1.011 | [1.009, 1.013] | $< 1\text{e-}16$ |
| resp_rate  | 1.021 | [1.019, 1.023] | $< 1\text{e-}16$ |
| map        | 0.988 | [0.986, 0.990] | $< 1\text{e-}16$ |
| sofa_total | 0.762 | [0.748, 0.777] | $< 1\text{e-}16$ |
| sirs       | 1.150 | [1.104, 1.198] | $2.5\text{e-}11$ |

Table 1 shows **SIRS** and the **vital signs** carry the hazard, while a higher **single-timepoint SOFA** is associated with a *lower* hazard of *incident* sepsis (HR 0.76): patients with organ dysfunction but no evolving infection at hour 6 tend not to cross the Sepsis-3 threshold later. Crucially, the global test of the **proportional-hazards assumption fails** ( $p < 1\text{e-}16$ ) — the covariate effects **change over time**, which a baseline Cox cannot express and which motivates a time-varying model.

## 4.1 Time-varying Cox

Using the hourly panel as a counting process, covariates are allowed to **update every hour** (start–stop intervals), so the model uses each patient’s *current* physiology rather than a fixed baseline.

Table 2: Time-varying Cox: hazard of onset from hour-updated physiology (sampled stays).

| Term       | HR    | 95% CI         | p       |
|------------|-------|----------------|---------|
| sofa_total | 0.801 | [0.729, 0.880] | 3.6e-06 |
| hr         | 1.019 | [1.010, 1.028] | 1.4e-05 |
| map        | 0.993 | [0.982, 1.004] | 0.21    |
| lactate    | 0.991 | [0.811, 1.211] | 0.93    |

Table 2 updates every covariate hour by hour. The **current heart rate** is positively associated with imminent onset (HR 1.019 per bpm), whereas the **single SOFA value stays inversely associated** (HR 0.80): organ dysfunction without an evolving infection does not progress to Sepsis-3, whether read at admission or hour-by-hour. The lesson is twofold — allowing covariates to vary in time is what rescues the violated proportional-hazards assumption, and the predictive signal lives in **physiological dynamics (notably heart rate)** rather than in a severity snapshot, the same conclusion the discrimination analysis reaches. (With 161 onset events in the sampled counting process, these hourly hazards are directional rather than precise.)

## 5 Competing Risks

Death and discharge **compete** with sepsis onset: a patient who dies sepsis-free can never be observed to develop sepsis, so treating those as ordinary censoring overstates incidence. A **Fine–Gray** model targets the cumulative incidence of onset in the presence of the competing event.

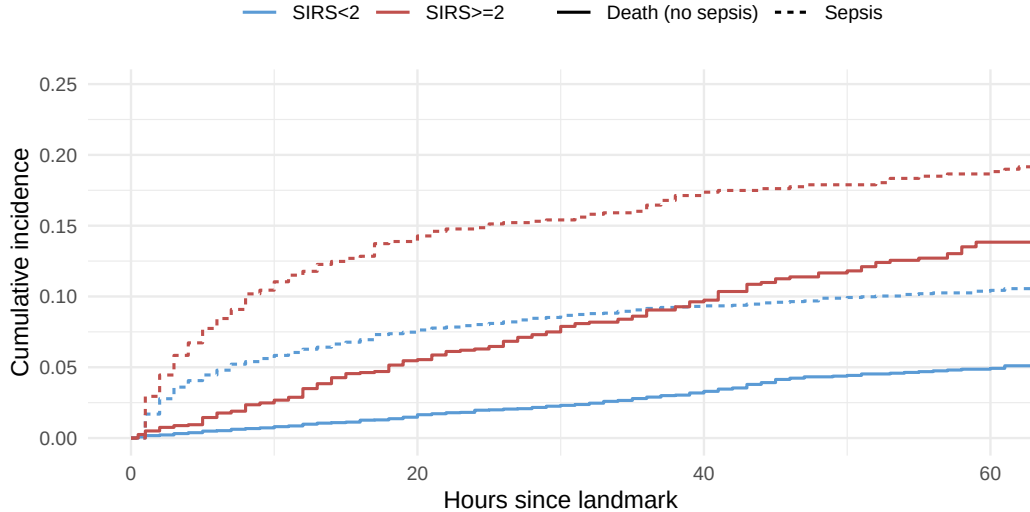


Figure 3: Cumulative incidence functions from the landmark, accounting for death without sepsis as a competing risk, by SIRS status.

Figure 3 shows onset incidence (solid) rising faster in the  $\text{SIRS} \geq 2$  group, while the competing risk of death without sepsis (dashed) is modest over the horizon. The Fine–Gray subdistribution hazard ratio

for SIRS is **1.49**, confirming SIRS raises the *cumulative* likelihood of onset even after allowing for the competing event.

## 6 Temporal Structure: Trajectories and Autocorrelation

The cross-sectional report’s HMM disclaimer noted that stays “are not a true time series.” Within a patient, however, the hourly measurements emphatically *are*.

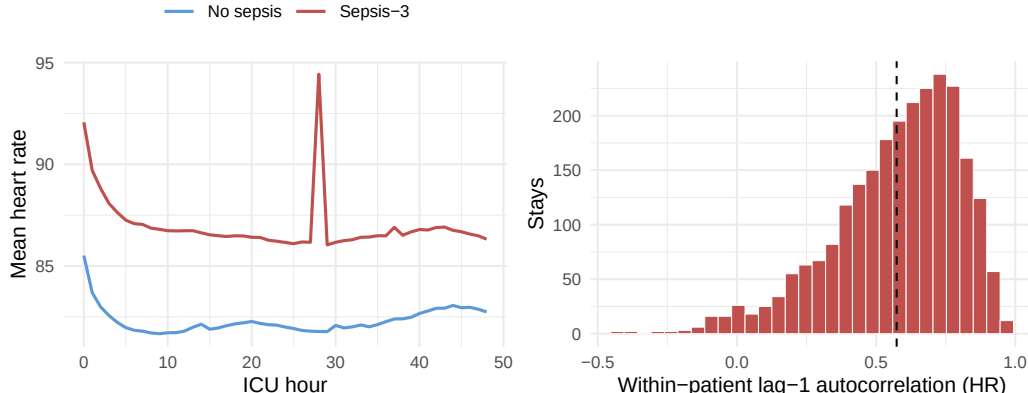


Figure 4: Left: mean hourly heart rate and SOFA by sepsis status (aligned to admission) — septic patients run higher and diverge. Right: distribution of within-patient lag-1 autocorrelation of heart rate; values well above zero show strong hour-to-hour dependence.

Figure 4 shows septic patients running at a higher mean heart rate throughout, and a mean within-patient lag-1 autocorrelation of heart rate of **0.57** — strong serial dependence. This dependence is the statistical foundation the static analysis discarded and that sequence models are built to exploit.

## 7 Within-Patient Hidden Markov Model

Here the HMM is applied as intended: to each patient’s **hourly heart-rate sequence**, with a latent state that persists and transitions over time. A three-state Gaussian HMM is fitted jointly across a sample of stays (`ntimes` = each stay’s length).

Table 3: Three-state HR HMM: emission means, diagonal (persistence) of the transition matrix, and the fraction of hours in each state that are labelled septic.

| State    | HR mean | Self-persistence | Septic-hour rate |
|----------|---------|------------------|------------------|
| 1 (low)  | 62.6    | 0.97             | 33.0%            |
| 2 (mid)  | 80.8    | 0.95             | 37.0%            |
| 3 (high) | 102.6   | 0.96             | 45.3%            |

Table 3 shows the three states order into low, medium and high heart-rate regimes with **strong self-persistence** (diagonal transition probabilities around 0.96) — genuine temporal stickiness, unlike the artefactual diagonal the cross-sectional HMM produced on stay *order*. The high-HR state also carries the highest proportion of septic hours, so the latent state tracks physiological deterioration. This is the HMM doing real work: modelling how a patient moves between hidden physiologic regimes over time.

## 8 Time-Dependent Discrimination and Clinical Baselines

Finally we establish the **bar** a real-time model must clear: how well do the standard streaming scores predict incident sepsis, and does using the *trajectory* help?

Table 4: Discrimination for incident sepsis within 12h of the landmark (cross-validated for the multivariable models).

| Score                                | AUC (within 12h) |
|--------------------------------------|------------------|
| <b>Single clinical scores</b>        |                  |
| SOFA (hour 6)                        | 0.359            |
| qSOFA (hour 6)                       | 0.492            |
| SIRS (hour 6)                        | 0.586            |
| <b>Multivariable</b>                 |                  |
| Multivariable (static, 6h)           | 0.749            |
| Multivariable + 6h trajectory slopes | 0.755            |

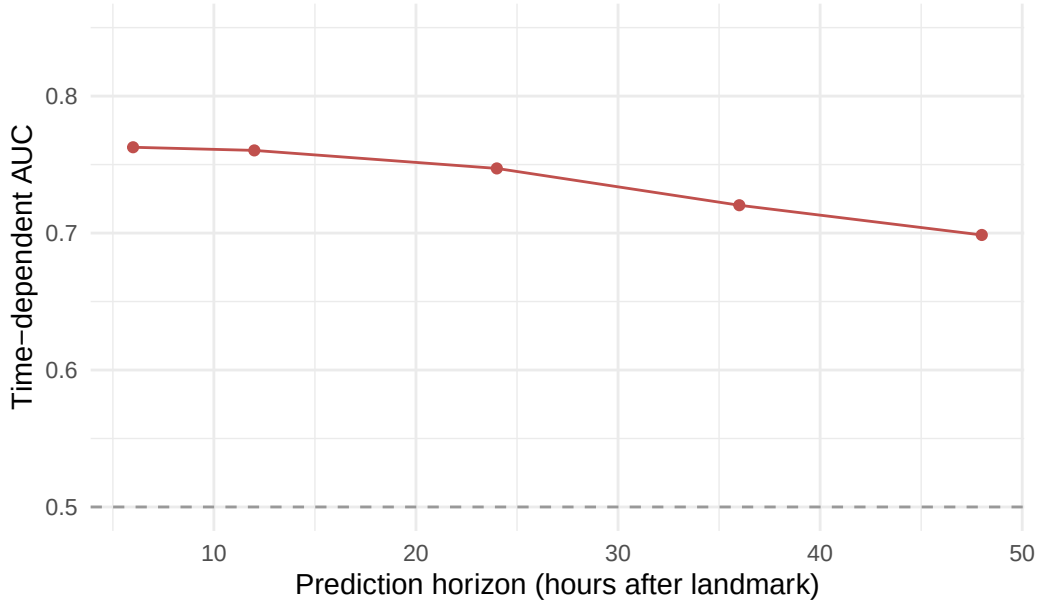


Figure 5: Time-dependent AUC for predicting sepsis onset over increasing horizons, using the cross-validated multivariable risk score.

Table 4 is the report’s punchline. **No single clinical score is useful** for incident sepsis — SOFA is even *inverse* (AUC 0.359), because organ dysfunction without evolving infection does not progress to Sepsis-3; SIRS is the best single score (AUC 0.586). But a **multivariable** combination of the same hourly measurements reaches AUC **0.749**, and adding **six-hour trajectory slopes** raises it further to **0.755** (DeLong  $p$  3.5e-07). Figure 5 shows this discrimination holds across horizons. The temporal features add real, statistically significant signal — direct evidence that the trajectory carries information a snapshot does not.

## 9 Conclusion and Path to a Real-Time Model

Reframing the problem longitudinally changed both the methods and the findings.

- **Prevalent vs incident.** A third of Sepsis-3 is present at admission and not predictable; the real task is the incident sepsis isolated by the hour-6 **landmark** (9.7% of at-risk stays).
- **The signal is in the dynamics, not the snapshot.** A baseline Cox violated proportional hazards; a **time-varying Cox** located the hazard in *current heart rate* rather than a SOFA severity score (which stays inversely related to incident onset). **Autocorrelation** (0.57) and a persistent-state **HMM** confirmed genuine within-patient temporal structure that the cross-sectional analysis had to discard.
- **A bar for the model.** Single streaming scores are weak or inverse; a multivariable hourly model reaches AUC **0.749** and trajectory features push it to **0.755**. Any real-time model should be judged against these, with time-dependent AUC and competing-risk-aware, leakage-safe, patient-grouped evaluation.

**Toward the real-time model.** The labelled hourly panel, onset times, landmark design, competing-risk framing, and streaming baselines assembled here are the direct inputs to the real-time statistical model built next (scripts 10-19): a dynamic **landmark supermodel** — fit entirely with penalized logistic regression, GEE, and time-varying Cox, no gradient boosting or neural sequence models — that emits an hourly sepsis risk by stacking predictions across a grid of landmark times. The statistics in this report specify the target, the evaluation, and the minimum performance that model must exceed.

**Limitations.** Onset depends on clinician behaviour (when cultures and antibiotics are ordered), so incident “onset” partly reflects *recognition* timing; the SOFA baseline is assumed zero; qSOFA uses MAP as a proxy for systolic pressure; and the time-varying Cox and HMM use a sampled subset for tractability. These are revisited as the modelling matures.

## Acknowledgment

This longitudinal analysis extends the cross-sectional study to the project’s real objective — real-time prediction — using the **PAIR framework** for responsible AI use: the problem framing, survival and time-series methods, and their honest (including negative) interpretation were critically reviewed against primary sources.

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